# Network Manager Release Notes

Version 2.1

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# 1: Introduction

This document provides a summary of the major changes in this release of Network Manager for Telecoms.

## 1.1 Intended Audience

This document is intended for Administrators and Developers with a detailed technical knowledge of IQGeo Platform and utilities.

# 2: Installation

The following changes relate to installation and upgrade.

## 2.1 Prerequisites

| Component | Version | Comments |
|---|---|---|
| IQGeo Platform | 6.2 | Mandatory |
| IQGeo Workflow Module | 6.2 | Mandatory |
| IQGeo Anywhere | 6.2.0 (or higher) | Optional |

All components must include any released patches.

Note that the IQGeo Workflow Module is no longer optional. See *Installation Process Simplified* on page 7 for more details.

## 2.2 Installing the Software

1. Uncompress the software download and copy the **comms** folder into:

   ```
   <myWorld_install_dir>/WebApps/myworldapp/modules
   ```

2. Download the module's JavaScript dependencies:

   ```
   myw_product fetch node_modules
   ```

3. Rebuild your JavaScript bundles:

   ```
   myw_product build applications
   myw_product build config
   ```

## 2.3 Creating a New Database

If you do not already have an IQGeo database, or that database does not yet have the Network Manager tables installed, you will need to:

- Install the Network Manager tables.
- Configure the network object types for your installation.

These processes are covered in detail in the *Installation and Configuration Guide*.

## 2.4 Upgrading an Existing Database

If you already have a Network Manager database from a previous release, you will need to:

- Upgrade your database.
- Perhaps perform some post-upgrade configuration changes.

These processes are covered in detail in *Upgrading* on page 19.

# 3: Functional Changes

## 3.1 Installation Process Simplified

The JavaScript application definition shipped with the product (main.mywcom.js) now includes configuration of the workflow plugins and design toolbar. This removes the need to create a custom application definition in order to access this functionality.

The product now also incudes a design template model (see below). You can use this to add a simple design feature type, state model and forward view layers.

## 3.2 Template Model Componentised

The template model has been split into the following components:

| Model | Description |
|---|---|
| structures | Building, manhole, cabinet etc |
| routes | Overhead and underground routes |
| conduits | Simple conduits (ducts) |
| internals | Room, floor, rack, shelf, slot and splice_closure |
| fiber | Fiber equipment and cables |
| ftth | Fiber-to-the-home |
| designs | Design object and state model |

Each model can be installed separately. This allows you to choose which objects you add to your database.

The system now also supports supplying template models in separate modules. For details on available models contact IQGeo support.

## 3.3 Reporting Added

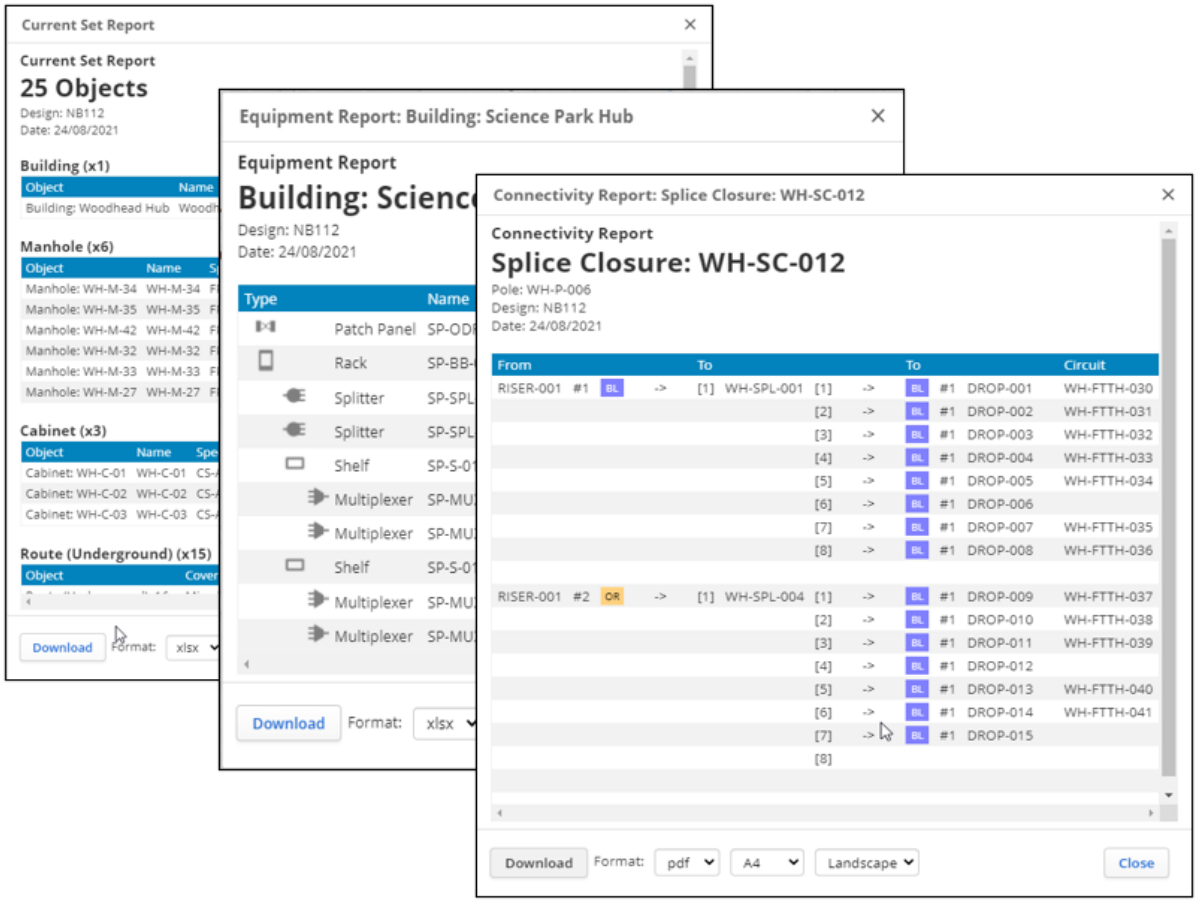



Figure 3-1: Reporting

The system now provides a variety of reports that can be download as standalone documents. These include:

- Equipment Report: Shows all equipment installed within a structure, with its specification and port usage.
- Connectivity Report: Shows splices and port connections within a structure, including fiber colours.
- Fiber Trace Report: Shows hops in a fiber trace, including port/fiber number and distance from start point.
- Current Set Report: Shows objects in the current selection, with their attributes.

- Validation Report: Shows validation errors and design rule violations detected by a Check Design operation.
- Bill of Materials (BOM) Report: Shows a summary of objects added in a design, with their specification and cost (where configured).
- All reports are downloadable in PDF, HTML, Excel and CSV formats. For more details, see the *User Guide*.

Some aspects of report content can be configured via database settings. You can also add your own reports, or modify existing ones, by implementing JavaScript Report classes and registering them with the ReportManager plugin. For more details, see the *Install and Configuration Guide*.

## 3.4 Design Import Added

Figure 3-2: Design Import

Network Manager now supports import of design data from external systems. The main features are:

- Supports import of attributes, connectivity and containment for all network object types.
- Converts coordinate system and units on-the-fly.
- Supports mapping of object types and attributes between network models.
- Can be initiated from command line tool or GUI.

The out-of-box system supports import and export in IQGeo Comms Data Interchange Format (CDIF format). You can extend the system to handle other formats by defining a Python DataImporter class. For details, see the *Developer Guide*.

For a definition of CDIF formats, see the *CDIF Specification*.

# 3.5 Design Changes GUI Enhanced

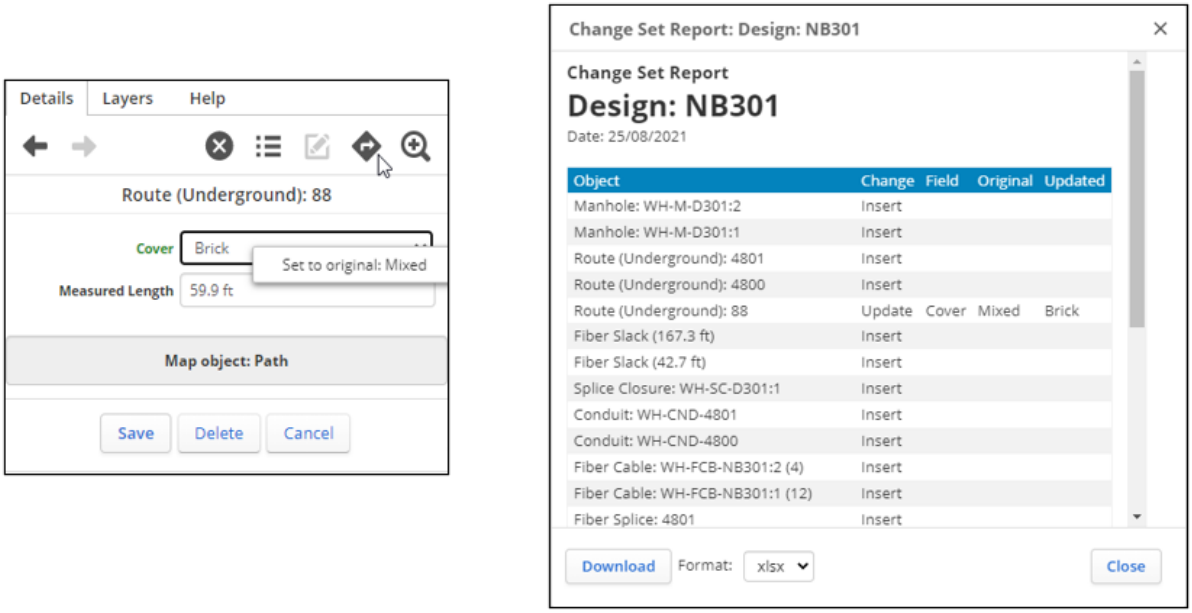



Figure 3-3: Design Changes GUI

The design Select Objects operation as been extended as follows:

- The Results pane now shows more details about the fields that have been changed (in hover text).
- Object viewers now highlight fields that have been updated (in green) and show the original value (hover text). Hovering over an updated geometry displays its original location on the map (green).
- Object editors now provide a mechanism for restoring fields to their original value (via context menu).
- The Change Set Report provides a list of the changes made within the design.
- The BOM Report provides a summary of the design changes, grouped by object type and specification.

For more details, see the *User Guide*.

## 3.6 Conflicts GUI Enhanced

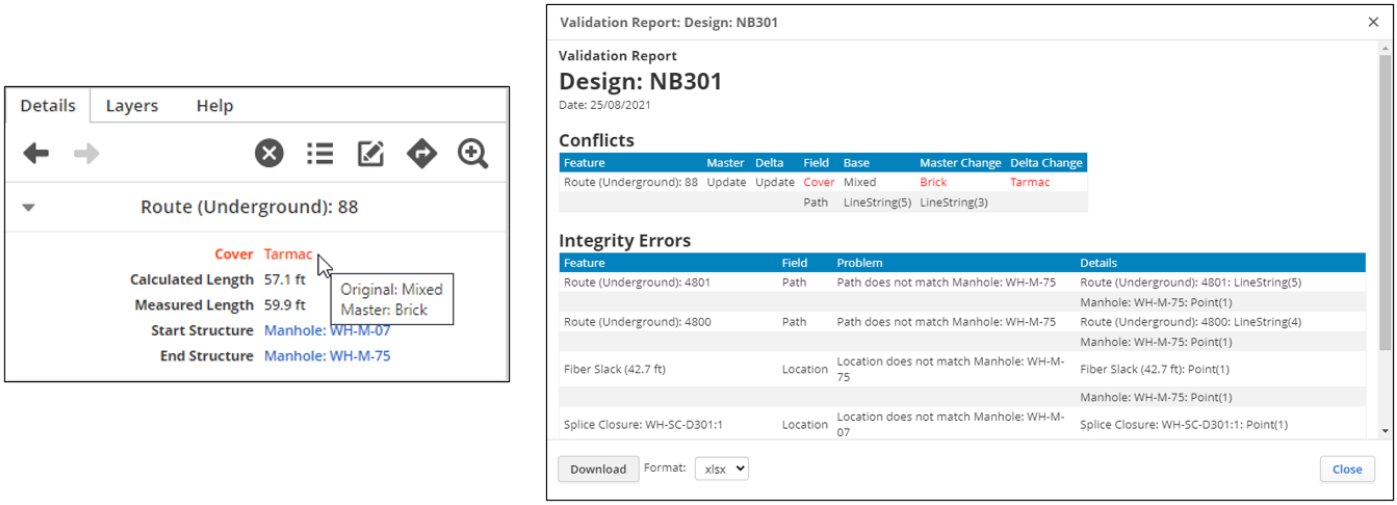



Figure 3-4: Design Changes GUI

The Check Design operation as been extended to provide more details on the validation errors and design rule violations it finds:

- The Results pane now shows details of the fields in conflict (in hover text).
- Object viewers now highlight fields that are in conflict (in red) and show the original and master values (hover text).
- Object editors now provide a mechanism for updating fields to the master value (via context menu).
- The Validation Report provides a summary of the problems found.

For more details, see the *User Guide*.

## 3.7 Configuration Process Simplified

The Comms Settings page has been enhanced to automatically update housing reference fields and network definitions when a feature type is added to a category. This greatly simplifies the process of adding new structure and equipment types to your data model.

For more details, see the *Installation and Configuration Guide*.

## 3.8 Command Line Tool Renamed and Extended

The command line tool has been modified as follows:

1. Name: The tool has been renamed as follows:

| Old Name | New Name |
|---|---|
| comms_model_util | comms_db |

This change is to reflect its additional functions (see below).

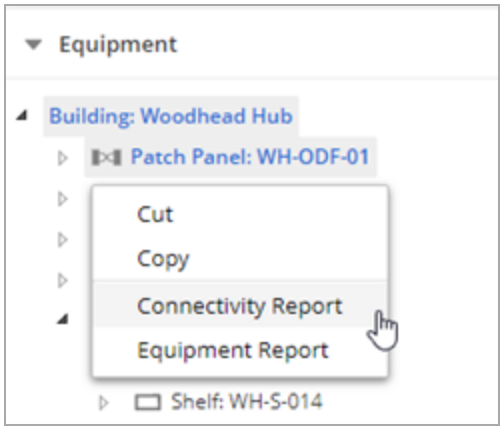
1. Operation install: Install accepts a new option --module. This is used when installing a template model from a separate module.
2. Operation validate: Validate now supports validation of feature data (as well as configuration). See --help for details.
3. Validate --warn: This flag is removed. To see configuration warnings use --verbosity=3 or higher.
4. New operations: The tool supports the following new operations:

| Operation | Description |
|---|---|
| list | Show contents of a data package |
| import | Import a data package |
| export | Dump data as a data package |

These operations are described in detail in *Installation and Configuration Guide*.

# 4: End-user GUI Changes

## 4.1 Equipment Tree

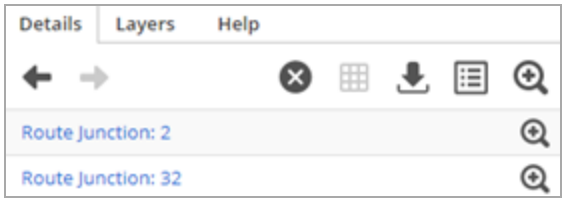



The **Equipment Tree** has been enhanced as follows:

- Splice nodes now show a count of the fibers they contain.
- The context menu for structure and equipment nodes includes new entries Equipment Report and Connectivity Report.

For more information on the equipment tree, see the *User Guide*.

## 4.2 Results Toolbar




The Results toolbar has been enhanced as follows:

- There is a new button Create Report that generates a report for the current result set.

Note that the type of report generated depends on the type of result currently displayed. For example, a query result will be displayed as a Current Set Report whereas a trace result will create a Fiber Trace Report.

For more details on the reporting, see the *User Guide*.

## 4.3 Design Toolbar

The Design Toolbar has been enhanced as follows:

- There is a new button Create bill-of-materials Report that generates a BOM.
- There is a new button to Import data that opens the Data Import dialog.

For more information, see the *User Guide*.

Note that Import data is not available in IQGeo Anywhere.

# 5: Administrator GUI Changes

## 5.1 Comms Settings Page

The Comms tab in the Configuration -> Settings page has been modified as follows:

1. Registering objects: The system now automatically adjusts select relationships and network definitions when you add a new network object type. This greatly simplifies the process of registering new structure and equipment types.
2. Feature type fields: Fields for selecting housings and other feature types are now restricted to valid choices using pulldown lists (rather than being free format text).
3. Help button: There is new button Help that launches the Comms documentation page.

## 5.2 Advanced Settings Page

There are new settings for configuring report content and defining import formats:

| Setting | Purpose |
|---|---|
| mywcom.equipmentReport | Defines extra columns in equipment report |
| mywcom.connectivityReport | Defines extra columns in connectivity report |
| mywcom.import_config.cdif | Defines mappings for import of CDIF packages |

For more information, see the *Installation and Configuration Guide*.

# 6: Data Model Changes

## 6.1 Spec Cost Fields Renamed

The following fields on specification records are migrated by the database upgrade:

| Old Field | New Field | Description |
|---|---|---|
| cost_unit | item_cost | Cost of purchasing one unit |
| cost_per_unit | item_cost | Cost of purchasing one unit |
| cost | item_cost | Cost of purchasing one unit |
| cost_length_unit | item_length | Length of one unit |
| | item_length_unit | Unit in which item_length is expressed |

This change corrects inconsistent naming in the 2.0 template model. Cost information is used when generating the BOM report.

If you have implemented your own BOM report you may need to update your code for this change.

## 6.2 Design Setting Modified

Setting mywcom.designs is converted to a JSON dictionary at this release (rather than an array). This brings it into line with the other category definition settings.

This change is unlikely to affect custom code.

## 6.3 Conduit Field Forward Removed

The database upgrade removes field forward from all conduit feature types. This field was used in release 1.1 to model reversal in continuous conduits. Conduit reversal was removed at release 2.0.

This change should not affect custom code since the field was always optional and, where present, should always have been set to true.

# 7: API Changes

## 7.1 JavaScript API

The supported JavaScript API has been modified as follows:

1. DisplayManagerPlugin: Method fiberColors() now returns a dictionary of FiberColor objects (rather than a zero-based array of structures). This makes accessing color information simpler.
2. FiberColorEngine: The color engine classes have been refactored. If you have added a custom fiber color engine will need to update your code.
3. ContainmentPlugin: The containment manager plugin is removed at this release. You should use methods structContent() and routeContent() on StructureManagerPlugin instead.
4. Reporting: There are new classes ReportManager, Report and ReportStream that can be used for implementing custom reports.

For more details on any of these changes contact IQGeo support.

## 7.2 REST API

There are no changes to the supported REST API at this release.

# 8: Documentation Changes

The documentation set has been updated for the changes described above. In addition:

- CDIF Specification: There is new document CDIF Specification that provides a definition of IQGeo Comms Data Interchange Format. It is intended as reference material for 3rd party data providers.

All documentation can be found in the *public/doc* directory.

The documentation is published in a responsive HTML format that adapts to the user's device and facilitates searching across guides.

# 9: Upgrading

This section provides an overview of the upgrade process for this release.

## 9.1 Upgrading Your Database

To upgrade from the previous release of Network Manager:

1. Apply all IQGeo Platform upgrades:

   ```
   myw_db <db_name> upgrade core 620
   ```

2. Apply the Network Manager upgrade:

   ```
   myw_db <db_name> upgrade comms 6221
   ```

3. Perform the post upgrade actions listed below.

## 9.2 Post-upgrade Actions

After upgrading your database, you should check the following:

1. Application JS definition: If your custom JavaScript application definition includes configuration of the DatabaseViewPlugin and CommsDeltaOwnerPlugin plugins these can now be removed. Configuration of these workflow-related classes is now included in the main.mywcom.js shipped with the product (see above).

2. Specification cost data: If you have configured cost information on your specification records you should check that it appears as expected after migration. If you plan you use the new BOM Report functionality you may also wish to replace currency acronyms (USD, GBP, etc) with symbols ($, £, etc).

   Note that the BOM Report requires a value in item_cost, item_length and item_length_unit in order to compute a cost for a linear object. If you wish to include costs on cables, conduits and routes in your BOM you must populate these three fields on your specifications.

You should also review any custom code for impact from the changes described in *API Changes* on page 17.

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